

chat pdf of lecture 19

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Use strictly and exclusively the provided context, which consists of lecture slides/PDF and relevant textbook excerpts. Every statement, equation, definition, derivation, and example must be traceable to the provided context. Do not introduce any external knowledge, additional examples, interpretations, or simplifications.

The lecture scribe must faithfully reconstruct the lecture content and include the following components whenever they appear in the provided context:

- Definitions and notation exactly as presented
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- Worked examples including all intermediate steps
- Logical progression and dependencies between concepts
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Ensure the final output is precise, structured, complete, and fully aligned with the provided lecture and textbook content.



Below is a **faithful, fully structured lecture scribe** reconstructed strictly from the provided lecture slides of **CSE400 Lecture 19: N-Variate Joint PDF/PMF/CDF**. Every definition, formula, derivation, and step is preserved exactly as presented in the material.

CSE400 – Fundamentals of Probability in Computing

Lecture 19: N-Variate Joint PDF/PMF/CDF

Source: Lecture slides  L19

1. Joint Distribution of N Random Variables

1.1 Definition and Representation

Definition

For random variables $X_1, X_2, \dots, X_N$, the vector form is:

$$\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Key Statements

- The joint distribution describes the probability behavior of all variables together.
- It can be represented using:
 - Joint PMF
 - Joint PDF
 - Joint CDF

2. Joint Gaussian Random Variables

2.1 Joint Gaussian PDF (Two Variables)

For two variables X and Y :

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y \sqrt{1-\rho^2}} \exp \left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X} \right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y} \right)^2 - 2\rho \left(\frac{x-\mu_X}{\sigma_X} \right) \left(\frac{y-\mu_Y}{\sigma_Y} \right) \right] \right)$$

Where

- μ_X, μ_Y : means
- σ_X, σ_Y : standard deviations
- ρ : Pearson correlation coefficient

2.2 General Formula

$$f_{XY}(x,y) = \frac{\exp \left\{ -\frac{1}{2(1-\rho_{XY}^2)} \left[\left(\frac{x-\mu_X}{\sigma_X} \right)^2 - 2\rho_{XY} \frac{(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \left(\frac{y-\mu_Y}{\sigma_Y} \right)^2 \right] \right\}}{2\pi\sigma_X\sigma_Y \sqrt{1-\rho_{XY}^2}}$$

3. Example 1: Joint Gaussian Variables

3.1 Given

Two random variables X and Y are jointly Gaussian with:

$$\mu_X = 2, \quad \sigma_X^2 = 1$$

$$\mu_Y = -3, \quad \sigma_Y^2 = 4$$

$$\text{cov}(X, Y) = -1$$

3.2 Problem Statement

Find:

- (a) Joint PDF $f_{XY}(x, y)$
- (b) Marginal PDFs $f_X(x)$ and $f_Y(y)$
- (c) $\Pr(X < 0)$ in terms of Q-function

4. Solution (a): Joint PDF Construction

4.1 Step 1: Correlation Coefficient

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

From given variances:

$$\sigma_X = \sqrt{1} = 1, \quad \sigma_Y = \sqrt{4} = 2$$

Substitute:

$$\rho_{XY} = \frac{-1}{(1)(2)} = -\frac{1}{2}$$

4.2 Step 2: Substitute into General Formula

Substitute:

$$\mu_X = 2, \sigma_X = 1, \mu_Y = -3, \sigma_Y = 2, \rho_{XY} = -0.5$$

4.3 Final Joint PDF

$$f_{XY}(x, y) = \frac{1}{2\pi \cdot 3} \exp \left\{ -\frac{2}{3} \left[(x-2)^2 + \frac{(x-2)(y+3)}{2} + \frac{(y+3)^2}{4} \right] \right\}$$

5. Solution (b): Marginal PDFs

5.1 Definition

Marginal PDFs are obtained by integrating the joint PDF:

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx$$

5.2 Key Property

Because X and Y are jointly Gaussian:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

5.3 Final Marginal PDFs

$$f_X(x) = \frac{1}{2\pi} \exp\left(-\frac{(x-2)^2}{2}\right)$$

$$f_Y(y) = \frac{1}{2\pi} \exp\left(-\frac{(y+3)^2}{8}\right)$$

6. Solution (c): CDF and Standardization

6.1 Step 1: Identify the CDF

$$F_X(x) = \Pr(X \leq x)$$

Thus:

$$F_X(0) = \Pr(X < 0)$$

6.2 Step 2: Standardization

$$Z = \frac{X - \mu_X}{\sigma_X}$$

Given:

$$X \sim \mathcal{N}(2, 1)$$

$$\begin{aligned} \Pr(X < 0) &= \Pr\left(\frac{X - 2}{1} < \frac{0 - 2}{1}\right) \\ &= \Pr(Z < -2) \\ &= \Phi(-2) \end{aligned}$$

6.3 Step 3: Symmetry Property

$$\Phi(-x) = 1 - \Phi(x) = \Pr(Z > x)$$

Thus:

$$\Pr(Z < -2) = \Pr(Z > 2)$$

6.4 Step 4: Q-function Relation

$$Q(x) = \Pr(Z > x)$$

Final Answer

$$\Pr(X < 0) = Q(2)$$

7. Application: Large MIMO System

System Setup

- Wireless communication system
- M transmit antennas
- N receive antennas

This application demonstrates:

- Joint modeling of multiple random variables
- Use of joint distributions in communication systems

(Reference diagram shown in lecture slide)  L19

8. Summary of Key Concepts (From Lecture Flow)

- Joint distribution of N random variables represented in vector form
 - Joint PDF/PMF/CDF describe combined probabilistic behavior
 - Joint Gaussian distribution fully characterized by:
 - Means
 - Variances
 - Correlation coefficient
 - Marginals of jointly Gaussian variables are Gaussian
 - Standardization converts variables to $\mathcal{N}(0, 1)$
 - Q-function represents tail probabilities
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